Training-Free Partial-Order Control for Discrete-Diffusion POI Check-in Generation

Anonymous Author(s)

Abstract

Location-based social networks (LBSNs) are a core source of web-scale human-mobility data, yet sharing raw check-in traces is privacy-sensitive, motivating realistic synthetic generation. Beyond statistical realism, many downstream web applications need control over the *semantic order* of activities—e.g., leisure before dining—a sample-level, cross-position requirement that single-step conditioning cannot express. Enforcing such constraints is commonly assumed to cost distributional fidelity; we show this trade-off is largely *separable*. We present PCDG, a *training-free* control stack for a frozen conditional joint temporal + category/POI discrete-diffusion generator. PCDG compiles category partial-order constraints into differentiable energies, projects the category-position logits through a KL-preserving augmented-Lagrangian optimization in the late reverse steps, reduces residual category–POI inconsistency via a category-consistent decoder, and restores spatial realism through a distribution-aware same-category geometry refinement that leaves every constraint metric unchanged *by construction*. Across two cities under an out-of-distribution split and three sampling seeds, PCDG cuts the strict ordering-violation rate from 92.89% to 4.04% on Istanbul and from 89.45% to 11.28% on New York, raising pairwise ordering coverage by 6.5× and 4.2×, while *improving* all six fidelity divergences over the unconstrained backbone on Istanbul and four of six on New York. On Istanbul it surpasses a *fully retrained* partial-order conditional model on all five constraint metrics and five of six fidelity divergences, without retraining anything. Stage-wise ablations localize where fidelity is lost and recovered across the three-stage stack.

CCS Concepts

• **Information systems** → **Location based services**; *Data mining*; • **Computing methodologies** → **Machine learning**; Neural networks; • **Human-centered computing** → *Social networks*.

Keywords

synthetic human mobility, discrete diffusion, partial-order constraints, training-free constrained sampling, POI check-in sequences

1 Introduction

Location-based social networks (LBSNs) connect users' online activities with visits to physical places, producing time-stamped check-in sequences that support human mobility modeling and POI recommendation [9, 25, 40]. Generating realistic check-in sequences supports behavioral simulation and the evaluation of location-based Web services [7, 10, 23, 42]. These synthetic traces also provide controllable inputs for Web applications such as recommender stress tests and service simulation, where preserving population-level mobility statistics is as important as satisfying a requested activity pattern. Relevance to the Web follows from this setting: the resulting traces support controllable simulation and stress testing of location-based services while avoiding the release of individual check-in logs.

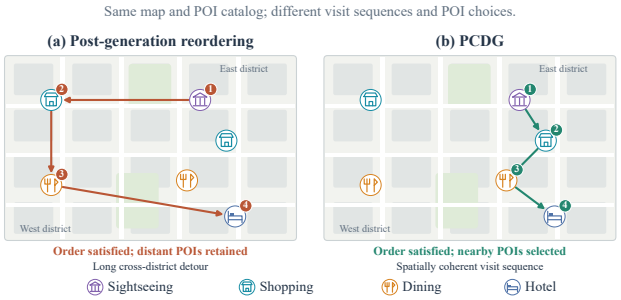


Figure 1: Illustrative activity-order control on a shared map. Both routes satisfy sightseeing before shopping. (a) Reordering retains sampled POIs and creates a cross-district detour; (b) PCDG selects nearby POIs within the same categories. These routes are illustrative, not measured samples or a shortest-path guarantee.

Activities in check-in sequences can be represented by POI categories, such as sightseeing, shopping, and dining [7, 23, 41]. Beyond statistical fidelity, itinerary planning and personalized tour-package generation require control over the relative order of activities. Users may specify sightseeing before shopping while leaving dining flexible, rather than prescribing a complete activity sequence. Partial orders capture these requirements without fixing the entire itinerary or the specific POIs. We therefore study partial-order-constrained POI check-in generation, where required activity categories must appear in the specified relative order.

Existing mobility generators model spatiotemporal patterns through neural networks, point processes, and diffusion models [7, 19]. Marionette [7] incorporates global and partial mobility contexts into fine-grained trajectory generation. Geo-Llama [19] generates check-in sequences with explicit visit constraints, while ControlTraj uses road topology and trip attributes to control trajectory generation [45]. However, these forms of control do not provide an explicit mechanism for preserving user-specified category-level partial orders in generated check-in sequences. Post-generation reordering offers a straightforward alternative, but it cannot introduce missing required categories and may disrupt spatial coherence. As Figure 1 illustrates, retaining sampled POIs while changing visit order can create long inter-area travel. Coherent generation therefore requires coordinating category precedence with POI selection, rather than rearranging an already generated itinerary.

Partial-order-constrained generation presents three challenges. **First, user-specified partial orders must be accommodated at generation time without retraining for each order.** Users

may request different combinations of relations that are unknown beforehand, making it impractical to anticipate all orders in a fixed training scheme. **Second, constraint satisfaction requires both the presence and correct ordering of the specified categories.** An ordering penalty alone admits trivial solutions that omit required categories. **Third, satisfying partial orders must preserve realistic spatial patterns.** Required categories and their relative order can change POI choices, travel distances, and visit patterns, so constraint satisfaction and spatial fidelity must be coordinated.

In response to these challenges, we introduce Partial-Order Controllable Diffusion Generation (PCDG), a training-free framework for partial-order- constrained POI check-in generation with a frozen conditional diffusion generator. PCDG expresses category precedence through a cross-position energy and couples it with an existence energy that penalizes missing categories. Sampling-time projection incorporates these energies into KL-anchored constrained optimization, discouraging both incorrect ordering and omission. Category-consistent decoding restricts supported POI choices to the sampled activity categories, connecting category-level control to the emitted POIs. After diffusion, category-preserving spatial refinement reselects POIs within their realized categories, improving spatial fidelity while leaving the realized category sequence and its constraints unchanged. On Istanbul and New York, PCDG reduces the strict ordering-violation rate from 92.89% to 4.04% and from 89.45% to 11.28%, respectively, without parameter retraining.

Our contributions are threefold:

- We formulate partial-order-constrained POI check-in generation, in which required activity categories must appear in a specified relative order without prescribing a complete itinerary.
- We propose PCDG, a training-free framework that combines joint order-and-existence projection, category-consistent POI decoding, and category-preserving spatial refinement to coordinate activity-order control with POI selection.
- Experiments on two real-world LBSN datasets demonstrate improved constraint satisfaction. Ablation and stage-wise analyses identify the complementary roles of the components and spatial discrepancies that can be reduced without altering the realized activity sequence.

2 Problem Definition

Definition 2.1 (POI check-in sequence). *A POI check-in sequence of length n is $s = (t_{1:n}, k_{1:n}, l_{1:n})$, where $t_{1:n}$ are arrival times, $k_i \in \mathcal{C} = \{1, \dots, C\}$ are activity categories, and $l_i \in \mathcal{L}$ are points of interest. Each POI l has fixed coordinates* $\mathrm{gps}(l)$ *and a catalog category* $\mathrm{cat}(l)$.

Definition 2.2 (Sequence-level context). *A sequence-level context is a tuple $\mathbf{c} = (\mathbf{c}^{(1)}, \dots, \mathbf{c}^{(R)})$, where each $\mathbf{c}^{(r)}$ is a coarse temporal or behavioral condition represented over fixed within-day time bins. The same context conditions the generation of the complete sequence.*

In the current backbone, $R = 6$ and the conditions are encoded over 24 daily time windows to obtain a sequence-level context embedding. These channels describe the background under which a sequence is generated; they do not specify POI identities, category precedence, or the partial-order matrix below. The generator's category token k_i and the emitted POI's category $\mathrm{cat}(l_i)$ are separate outputs, and the latter determines the realized activity.

Definition 2.3 (Category partial-order matrix). *A category partial-order matrix is a binary matrix $\mathbf{M} \in \{0,1\}^{C\times C}$, where $\mathbf{M}[a,b] = 1$ denotes the strict precedence requirement that category a precede category b, and $\mathbf{M}[a,b] = 0$ denotes no such requirement.*

Both user-specified partial-order constraints and a reference check-in sequence induce a matrix of this form. For a sequence s, let $I_s(a) = \{i : \mathrm{cat}(l_i) = a\}$ be the positions whose realized POI category is a. Its induced matrix is

$$\widehat{\mathbf{M}}(s)[a,b] = \begin{cases} 1, & I_s(a), I_s(b) \neq \varnothing \text{ and } \max I_s(a) < \min I_s(b), \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

For user-specified relations, $\mathbf{M}[a,b] = 1$ whenever a is required to precede b; for a reference sequence s_{ref}, we set $\mathbf{M} = \widehat{\mathbf{M}}(s_{\mathrm{ref}})$. A missing category or an inversion sets the corresponding induced entry to zero.

Definition 2.4 (Partial-order-constrained POI generation). *Given a frozen conditional generator $p_\theta(s \mid \mathbf{c})$ and a target matrix $\mathbf{M}$, sampling-time control produces*

$$\hat{s} \sim p_{\mathrm{PCDG}}(s \mid \mathbf{c}, \mathbf{M}), \qquad \mathbf{M}[a,b] \le \widehat{\mathbf{M}}(\hat{s})[a,b] \quad \forall a, b, \tag{2}$$

or, equivalently,

$$\mathbf{M} \odot \left(\mathbf{1}_{C\times C} - \widehat{\mathbf{M}}(\hat{s})\right) = 0.$$

The condition in (1) checks only relations required by $\mathbf{M}$; additional valid relations are allowed. A relation fails if either category is absent or their order is inverted. Since $\widehat{\mathbf{M}}$ uses $\mathrm{cat}(l_i)$, satisfaction is measured on realized POI categories. Unlike next-location prediction [9, 25, 40], the task generates a complete time-stamped, categorized, geolocated sequence under constraints coupling multiple, potentially non-adjacent positions.

We evaluate along two axes that are in tension. *Distributional fidelity* is measured by the Jensen–Shannon divergence (JSD) between generated and reference marginals of travel distance, centroid-based Radius, category-transition structure, daily location count, hourly category usage, and a global ranking statistic (G-RANK). *Constraint satisfaction* is measured by ordering-violation rates (OVR), pairwise and category coverage of the required relations, and an overall unsatisfied-relation rate (Unsat); full definitions are given in §?? and Appendix D. These measures jointly distinguish correct ordering from the disappearance of required activities.

3 Method

PCDG controls a frozen generator through sampling-time logit projection, category-consistent POI decoding, and terminal same-category refinement (Figure 2). The weights ρ, λ, β enter their objectives.

3.1 Backbone: conditional joint discrete diffusion

The current implementation uses a frozen generator whose discrete branch follows the Marionette-style diffusion point process

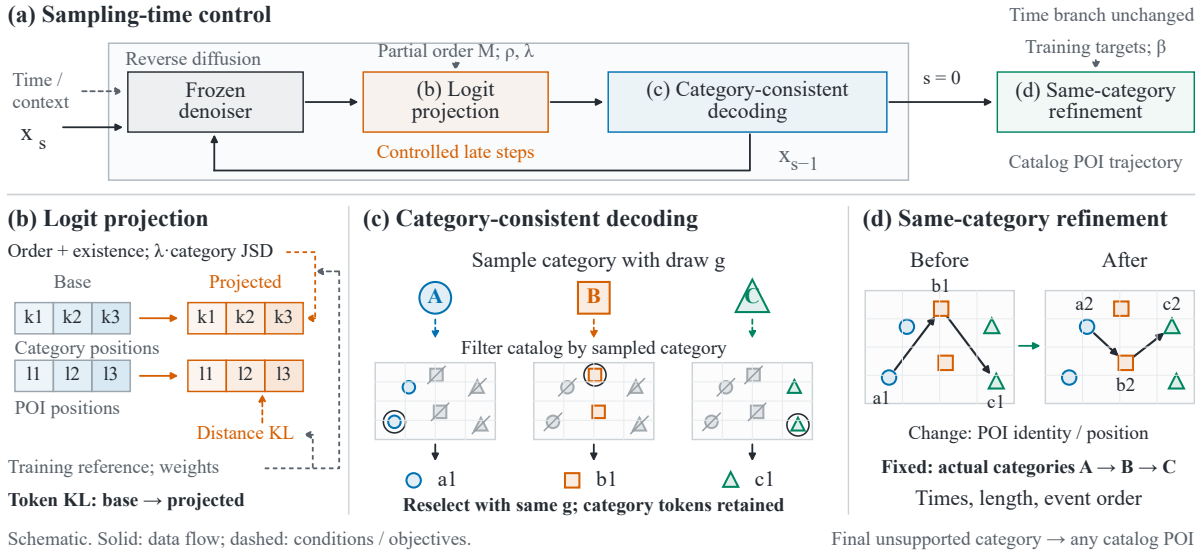


Figure 2: PCDG (schematic). (a) Frozen reverse loop and terminal refinement. (b) Category order/existence energies and POI distance KL, anchored by base-to-projected token KL. (c) Category-filtered POI reselection reuses Gumbel noise. (d) Same-frame refinement preserves actual categories, times, length, and order. Dashed edges carry conditions/objectives. Intermediate steps allow POI masks and leave unresolved categories unfiltered. Unsupported final categories allow any catalog POI, permitting nonzero mismatch.

for arrival times and conditional discrete diffusion for categories and POIs [7]. Its discrete branch represents a sequence as $\mathbf{x} = [\text{START}, k_{1:n}, \text{SEP}, l_{1:n}, \text{END}]$, with a *category mask* and a *POI mask* identifying the two segments. Categories and POIs use separate absorbing-state transition matrices [1]; a Transformer denoiser with an $\mathbf{x}_0$-parameterization predicts clean-token distributions, from which the backbone constructs the reverse-step distribution $p(\mathbf{x}_{s-1} \mid \mathbf{x}_s)$. We denote its per-position log-probabilities by $\mathbf{z} \in \mathbb{R}^{V \times L}$, with vocabulary size V and L positions. Projection acts on these reverse-step logits, not directly on the denoiser's clean-token predictions. PCDG keeps the backbone frozen and changes sampling distributions and terminal POI assignments.

3.2 Partial-order projection in the reverse process

Given the sample-level matrix $\mathbf{M}$, we parse the required pairs $\{(a,b) : \mathbf{M}[a,b] > \frac{1}{2}\}$. At each reverse step, temperature-T softmax probabilities (optionally with Gumbel noise $\mathbf{g}_l$) relax the categorical choices:

$$\boldsymbol{\pi}_l = \text{softmax}\big((\mathbf{z}_l + \mathbf{g}_l)/T\big) \in \Delta^{V-1}, \qquad l = 1, \dots, L. \tag{3}$$

We restrict attention to category positions $\mathcal{P}_k$ (via the category mask) and to the category slice of the vocabulary, writing $\pi_l^{\text{cat}}(a)$ for the relaxed probability that position $l \in \mathcal{P}_k$ takes category $a \in C$. For a constraint (a,b) let $P_l^A = \pi_l^{\text{cat}}(a)$ and $P_l^B = \pi_l^{\text{cat}}(b)$. We define two energies:

$$g^{\text{ord}}_{(a,b)} = \sum_{l \in \mathcal{P}_k} \Big(\textstyle\sum_{l' < l} P^B_{l'} \Big) P^A_l, \tag{4}$$

$$g^{\text{ex}}_{(a,b)} = \text{ReLU}\Big(1 - \textstyle\sum_l P^A_l\Big) + \text{ReLU}\Big(1 - \textstyle\sum_l P^B_l\Big). \tag{5}$$

The order energy (3) sums relaxed b-before-a pair mass; for a one-hot assignment, zero energy means no inversion. The existence energy (4) penalizes an expected count below one for either category, preventing omission-based satisfaction. Both energies use category positions only; POI logits do not enter these terms.

Let $\mathbf{y}^{(0)} = \mathbf{z}$ be the detached backbone logits and $\mathbf{y}$ a trainable copy. At controlled steps we reduce constraint violations while penalizing deviations from the model distribution:

$$\begin{aligned} &\min_{\mathbf{y}} D_{\text{KL}}\big(\text{softmax}(\mathbf{y}^{(0)}) \,\big\|\, \text{softmax}(\mathbf{y})\big) \\ &\text{s.t. } g^{\text{ord}}_{(a,b)} \le \tau, \quad g^{\text{ex}}_{(a,b)} \le \tau \quad \forall (a,b), \end{aligned} \tag{6}$$

where the KL direction is base-to-projected. We optimize the augmented-Lagrangian relaxation with $\delta^\bullet = \text{ReLU}(g^\bullet - \tau)$,

$$\begin{aligned} \mathcal{L}(\mathbf{y}) = w_{\text{kl}} D_{\text{KL}} &+ \sum_{(a,b)} w_{\text{ord}} \Big(\lambda^{\text{ord}} \delta^{\text{ord}} + \tfrac{1}{2} \mu^{\text{ord}} (\delta^{\text{ord}})^2 \Big) \\ &+ \sum_{(a,b)} w_{\text{ex}} \Big(\lambda^{\text{ex}} \delta^{\text{ex}} + \tfrac{1}{2} \mu^{\text{ex}} (\delta^{\text{ex}})^2 \Big). \end{aligned} \tag{7}$$

An inner SGD loop updates active rows of $\mathbf{y}$ with rate η and gradient clipping. The outer loop computes $\hat{\delta}$ from the hard arg max assignment and updates dual variables,

$$\lambda \leftarrow \lambda + \mu\, \hat{\delta}, \qquad \mu \leftarrow \min(\alpha\, \mu,\ \mu_{\max}) \text{ if } \hat{\delta} > \tau_{\text{tol}}, \tag{8}$$

Reported runs use a fixed budget without early stopping. Batch KL is averaged over the full batch; order/existence penalties are summed over active rows and relations (Appendix A). Hard violations determine dual updates. Control runs every f steps in the final K reverse steps, replacing category logits with projected values and, when enabled, POI logits with distance-adjusted values. Joint sampling produces the next state; the temporal branch is unchanged. Disabling control recovers the backbone. Hard POI-support masking belongs to the subsequent decoder. Algorithm 1 summarizes this controlled step.

3.3 Distributional regularization and calibration

To limit spatial drift, a batch-level distance KL matches the training population's route-length distribution. Cumulative Haversine lengths d from training sequences with at least two POIs form a fixed-bin histogram H_{train} over $\log(1+d)$ (the reference is content-hashed and cached, never stored in the model). For each POI position, the backbone's top-κ candidates define P straight-through Gumbel-softmax routes; exact distances are used forward and a bilinear surrogate backward to form $H_{\text{proj}}(\mathbf{y})$:

$$w_{\text{dist}}\, D_{\text{KL}}\big(H_{\text{train}} \,\|\, H_{\text{proj}}(\mathbf{y})\big) \tag{9}$$

to the objective (6). This term acts on POI logits and matches a batch distribution rather than imposing a per-sequence constraint. It applies only to sequences with at least two POIs, uses an independent random stream, and has one shared weight across cities. The weight is not tuned per city and is not guaranteed to improve every fidelity metric; its effect is measured by the No Distance KL ablation (§5).

Distribution correction uses the training reference pool to balance constraint strength and fidelity. The coefficient $\rho \in (0,1]$ scales the constraint penalty, while $\lambda \geq 0$ weights $\mathcal{L}_{\text{cat}} = \frac{1}{2}(\text{JSD}_{\text{hour}\times\text{cat}} + \text{JSD}_{\text{transition}})$ computed from expected counts. The POI-side coefficient $\beta \geq 0$ adds $\frac{1}{2}(\text{JSD}_{\text{top1000}} + \text{JSD}_{\text{DailyLoc}})$; top-1000 visitation aligns POI identities across the generated/reference union. Pre-refinement samples are shared across β settings. The selected configuration is $(\rho, \lambda, \beta) = (0.5, 0, 1)$ (Appendix B). The resulting projection objective is

$$\begin{aligned}\mathcal{L}_{\text{bal}} = w_{\text{kl}}\, D_{\text{KL}} + \rho \sum_{(a,b)} \text{AL}(g^{\text{ord}}, g^{\text{ex}}) + w_{\text{dist}}\, D_{\text{KL}}(H_{\text{train}} \| H_{\text{proj}}) \\ + \lambda\, \mathcal{L}_{\text{cat}} + \beta\, \tfrac{1}{2}(\text{JSD}_{\text{top1000}} + \text{JSD}_{\text{DailyLoc}}).\end{aligned} \tag{10}$$

3.4 Category-consistent POI decoding

The category and POI segments are sampled at separate positions. A correctly ordered category-token sequence can therefore yield POIs with different actual categories. Since constraint metrics use $\text{cat}(l)$ from the catalog, projection of category tokens alone does not ensure activity-order control.

The decoder couples the segments within each reverse step. Let $\widetilde{\mathbf{z}}$ be the projected logits $\mathbf{y}$, or $\mathbf{z}$ at an uncontrolled step. With Gumbel draw $\mathbf{g}$, first sample the full token string from these logits, fixing category token $\hat{c}_l$ aligned with each POI position. Then restrict its POI support and reselect using the *same* draw:

$$\hat{x}_l^{\text{cc}} = \arg\max_{v \in S_l} (\widetilde{z}_{lv} + g_{lv}), \quad S_l = \{v : \text{cat}(v) = \hat{c}_l\}. \tag{11}$$

For a supported category, logits outside S_l are set to $-\infty$, not a large finite penalty. Category positions are untouched, and the sampler consumes exactly one draw of the original shape per step. At intermediate steps the POI mask token remains admissible; unsupported category positions are unrestricted. At the final step, an unsupported category falls back to *any catalog POI*. Thus the implementation retains a small category–POI mismatch fraction, 0.0034%–0.0219% across the six full-stack runs, rather than eliminating it. The decoder introduces no parameters or logit prior.

3.5 Terminal same-category geometry refinement

The chosen POIs determine coordinates and can shift spatial marginals (§5.4). After diffusion, we therefore re-select each POI *within its own realized category*.

For every POI position we build a candidate set of at most κ_g=32 POIs sharing the realized category: the originally sampled POI, the backbone's highest-scoring candidates, the POIs closest to the generated trajectory's own centroid, and, in the distribution-corrected variant, the most frequently visited POIs of the training reference; duplicates are removed, the set is filled by model score, and ties are broken by catalog id so the construction is deterministic. Stage (d) uses the original POI, top-15 by score and nearest-16 to the centroid; stage (e) uses the original, top-11, nearest-12, and the 8 most visited training POIs. Scores come from one frozen, untruncated t=0 evaluation of the backbone; reference GPS from the test split never enters. Writing $\mathbf{w}$ for a relaxed assignment over candidates, we minimize

$$\begin{aligned}D_{\text{KL}}(q \,\|\, \mathbf{w})\; w_{\text{fid}} + w_{\text{d}}\, \text{JSD}\big(H^{\text{dist}}_{\text{train}}, H^{\text{dist}}(\mathbf{w})\big) \\ + w_{\text{r}}\, \text{JSD}\big(H^{\text{rad}}_{\text{train}}, H^{\text{rad}}(\mathbf{w})\big),\end{aligned} \tag{12}$$

where the two soft histograms are differentiable in $\mathbf{w}$ and the proposal q places half of its mass on the original POI and spreads the other half over the candidates by model score, so the fidelity term penalizes drifting away from the backbone's preference. Optimization uses 8 straight-through paths and a prequalified 200-step budget; Appendix A gives the settings. For the *hard* output, we retain the lowest-objective assignment among the original and 8 optimized paths. Empty, single-point and single-candidate positions, and batches in which the base projection never ran, are skipped.

Constraint neutrality. Because the refinement substitutes only POIs of the same realized category, together with their catalog coordinates, and leaves category tokens, the realized category sequence, event times, sequence lengths, event order and all conditions unchanged *per event*, every constraint-relevant statistic is invariant by construction: $\text{OVR}_{\text{strict}}$, OVR_{skip}, pair and category coverage, Unsat, *Category*, *CategoryTransition* and the category–POI mismatch rate. We verify this numerically on every test run, requiring agreement within 10^{-12}. This invariance permits spatial and popularity statistics to change, but does not guarantee a match to every reference distribution within the candidate sets. Refinement cannot repair the category-side divergences it preserves.

Controlled-step algorithm. Algorithm 1 summarizes one controlled step: relaxed energies update detached logits, hard assignments update duals, and category decoding shares its Gumbel draw

Algorithm 1 PCDG: one controlled reverse step

Require: reverse-step logits $\mathbf{z}$, constraints $\mathbf{M}$, position masks, training references

$\mathbf{y} \leftarrow \text{detach}(\mathbf{z})$; initialize duals
for $o = 1, \ldots, 10$ **do**
  for $j = 1, \ldots, 50$ **do**
    Relax $\mathbf{y}$ with Gumbel-softmax
    Compute order/existence and distribution terms
    $\mathcal{L} \leftarrow$ weighted objective in Eq. (11)
    SGD on active rows of $\mathbf{y}$; clip gradients
  end for
  Update duals from hard violations, Eq. (7)
end for
Sample category tokens; decode POIs with shared noise
return next discrete state (Eq. (9) and fallback)

with POI reselection. Geometry refinement follows the final reverse step and changes only same-category POI choices.

4 Experiments

We test constraint satisfaction, distributional fidelity, and the contributions of projection, decoding, and geometry refinement.

4.1 Experimental Setup

Datasets. New York has 3,160 training and 2,108 test sequences; Istanbul has 7,035 and 4,914. New York's sequence-level split reverses category- precedence preferences, but does not guarantee disjoint complete partial-order configurations. Istanbul uses the supplied historical OOD split, whose original construction provenance is not fully recoverable. Each test reference supplies an order matrix from actual category first/last positions (§3). Sampling-time methods share frozen city-specific backbones; PO-CFG is retrained.

Protocol. Order, existence, token-KL, and distance-KL weights are $(1, 5, 1, 1)$. Projection uses Gumbel temperature 3 every fourth step in the final 40 reverse steps: 10 outer by 50 inner iterations (5,000 updates per active batch), SGD rate 1, gradient clip 10, and no early stopping. Distance KL uses 8 routes, top-32 candidates and bins, and path temperature 1. Formal sampling uses 64-sequence FP32 batches with TF32 disabled. The full stack uses $(\rho, \lambda, \beta) = (0.5, 0, 1)$. Reported spreads are sample SDs across seeds 135398, 135399, 135400 at fixed checkpoints, not training variability or confidence intervals.

Baselines. JointGen is the unconstrained backbone. **PostSwap** stably reorders generated events according to $\mathbf{M}$. **EnergyGuide** takes one gradient step on order plus five times existence energy per controlled step, without projection's KL anchor or dual updates. **PO-CFG** is trained with partial-order conditioning for 1000 epochs and evaluated at classifier-free guidance scale 1; it is the only retrained baseline. Our variants are PCDG-P (projection), PCDG-PD (projection and category-consistent decoding), and PCDG (full).

Metrics. All tables use the archived evaluation-v2 protocol. Fidelity uses six JSDs: cumulative travel *Distance*, *Radius*, conditional *CategoryTransition*, distinct daily POI count (*DailyLoc*), hourly *Category* usage, and identity-aligned top-1000 POI visitation (*G-RANK*). Distance is conditional on generated/reference first and last categories

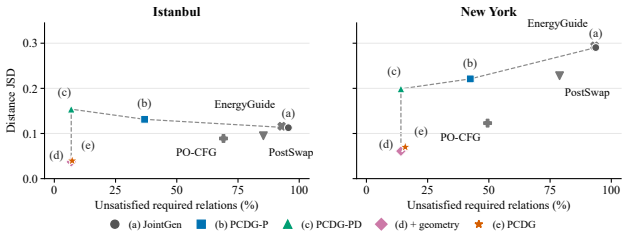


Figure 3: Unsat–Distance trade-off using the evaluation-v2 three-seed means in Tables 1 and 5; lower is better on both axes. Dashed paths connect (a)–(e); shared method/stage points are plotted once. Geometry (c)→(d) lowers Distance JSD without changing Unsat. Transition (d)→(e) changes three factors jointly, not a single regularizer.

matching; the eligible subset can differ by method. Radius is the square root of mean distance to the centroid, not standard radius of gyration. Appendix D specifies the fidelity statistics and filtering.

Constraint metrics use the catalog categories $\text{cat}(l)$ of emitted POIs, not the sampled category tokens. For reference sequence i, let R_i contain each distinct-category pair (a, b) satisfying $\max \text{pos}_i(a) < \min \text{pos}_i(b)$. Let $P_i \subseteq R_i$ contain pairs whose two categories both occur in the generated sequence; C_i is the set of categories involved in R_i and $\widehat{C}_i$ the generated category set. For a present pair, $v_i(a, b)$ is the number of inverted occurrence pairs divided by $n_a n_b$.

Strict OVR averages v_i over R_i, assigning 1 to missing-category pairs, then averages over sequences with nonempty R_i. Skip OVR averages v_i over P_i, then over sequences with nonempty P_i. Coverage and Unsat pool counts:

$$\text{PairCov} = \frac{\sum_i |P_i|}{\sum_i |R_i|}, \tag{13}$$

$$\text{CatCov} = \frac{\sum_i |C_i \cap \widehat{C}_i|}{\sum_i |C_i|}, \tag{14}$$

$$\text{Unsat} = \frac{\sum_i \sum_{(a,b)\in R_i} \mathbf{1}[(a, b) \notin P_i \ \vee \ v_i(a, b) > 0]}{\sum_i |R_i|}. \tag{15}$$

Unsat counts unsatisfied relations, not sequences; pair coverage requires both categories to occur, not correct order. Empty denominators are undefined, not zero; undefined OVR sequences are omitted from their respective means. Tables express these rates in percent.

4.2 Main results

Table 1 compares all methods on both cities.

Constraint satisfaction. The full PCDG reduces strict OVR from 92.89% to 4.04% on Istanbul and from 89.45% to 11.28% on New York, reductions of 88.85 and 78.17 percentage points. Unsat falls from 95.57% to 7.29% and from 93.64% to 15.86%, while pair coverage increases by 6.5× and 4.2×. All five constraint metrics improve over PO-CFG in both cities. PostSwap reaches zero skip OVR but leaves coverage unchanged, illustrating why skip OVR alone is insufficient. EnergyGuide gives much smaller gains.

Table 1: Method comparison (evaluation v2): fidelity JSDs (upper panel) and constraint metrics in percent (lower panel). Cells are mean ± sample SD over three sampling seeds at fixed checkpoints, not confidence intervals. Bold marks the best unrounded city-wise mean, not significance. PCDG-P: projection; PCDG-PD: adds decoding; PCDG: full stack. New York CategoryTransition and G-RANK remain above JointGen.

Istanbul						
Method	Distance ↓	Radius ↓	CategoryTransition ↓	DailyLoc ↓	Category ↓	G-RANK ↓
JointGen	0.113 ± 0.009	0.023 ± 0.002	0.040 ± 0.000	0.041 ± 0.001	0.065 ± 0.000	0.192 ± 0.002
PostSwap	0.095 ± 0.013	0.023 ± 0.002	0.036 ± 0.000	0.041 ± 0.001	0.065 ± 0.002	0.192 ± 0.002
EnergyGuide	0.116 ± 0.005	0.022 ± 0.002	0.042 ± 0.001	0.039 ± 0.002	0.064 ± 0.002	0.192 ± 0.000
PO-CFG (s=1)	0.089 ± 0.005	0.026 ± 0.002	0.036 ± 0.001	0.044 ± 0.002	0.063 ± 0.001	**0.179 ± 0.003**
PCDG-P (projection)	0.131 ± 0.004	0.052 ± 0.002	**0.016 ± 0.001**	0.037 ± 0.001	0.045 ± 0.002	0.182 ± 0.002
PCDG-PD (+decoding)	0.154 ± 0.003	0.089 ± 0.002	0.024 ± 0.001	0.030 ± 0.001	0.042 ± 0.001	0.181 ± 0.002
PCDG (full)	**0.040 ± 0.004**	**0.017 ± 0.001**	0.024 ± 0.001	**0.025 ± 0.001**	**0.042 ± 0.002**	0.180 ± 0.002
New York						
Method	Distance ↓	Radius ↓	CategoryTransition ↓	DailyLoc ↓	Category ↓	G-RANK ↓
JointGen	0.290 ± 0.016	0.110 ± 0.007	0.007 ± 0.001	0.041 ± 0.004	0.018 ± 0.000	0.161 ± 0.001
PostSwap	0.228 ± 0.045	0.110 ± 0.007	0.007 ± 0.001	0.041 ± 0.004	0.017 ± 0.001	0.161 ± 0.001
EnergyGuide	0.294 ± 0.006	0.111 ± 0.006	0.007 ± 0.001	0.041 ± 0.003	0.018 ± 0.000	0.162 ± 0.001
PO-CFG (s=1)	0.123 ± 0.016	0.082 ± 0.003	**0.005 ± 0.002**	0.050 ± 0.001	**0.012 ± 0.001**	**0.152 ± 0.002**
PCDG-P (projection)	0.221 ± 0.022	0.126 ± 0.005	0.013 ± 0.000	0.035 ± 0.004	0.017 ± 0.001	0.194 ± 0.004
PCDG-PD (+decoding)	0.199 ± 0.020	0.168 ± 0.004	0.018 ± 0.001	0.034 ± 0.002	0.018 ± 0.001	0.191 ± 0.006
PCDG (full)	**0.070 ± 0.011**	**0.037 ± 0.004**	0.018 ± 0.000	**0.030 ± 0.001**	0.018 ± 0.001	0.191 ± 0.006

Istanbul					
Method	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
JointGen	92.890 ± 0.080	48.050 ± 0.740	14.630 ± 0.190	43.190 ± 0.200	95.570 ± 0.150
PostSwap	86.100 ± 0.150	**0.000 ± 0.000**	14.630 ± 0.190	43.190 ± 0.200	85.370 ± 0.190
EnergyGuide	90.660 ± 0.060	44.760 ± 0.580	19.170 ± 0.430	47.060 ± 0.250	92.890 ± 0.160
PO-CFG (s=1)	63.450 ± 0.230	19.090 ± 0.540	43.720 ± 0.350	66.000 ± 0.230	69.070 ± 0.450
PCDG-P (projection)	32.800 ± 0.500	9.120 ± 0.100	70.990 ± 0.080	85.360 ± 0.150	36.860 ± 0.370
PCDG-PD (+decoding)	**3.720 ± 0.130**	1.760 ± 0.100	**94.740 ± 0.160**	**97.810 ± 0.070**	**6.900 ± 0.180**
PCDG (full)	4.040 ± 0.200	2.020 ± 0.150	94.680 ± 0.150	97.750 ± 0.100	7.290 ± 0.200
New York					
Method	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
JointGen	89.450 ± 0.200	52.920 ± 0.860	21.130 ± 0.150	46.750 ± 0.120	93.640 ± 0.260
PostSwap	78.220 ± 0.480	**0.000 ± 0.000**	21.130 ± 0.150	46.750 ± 0.120	78.870 ± 0.150
EnergyGuide	89.170 ± 0.140	51.590 ± 0.880	21.170 ± 0.190	47.140 ± 0.140	93.200 ± 0.390
PO-CFG (s=1)	41.910 ± 1.370	8.970 ± 0.880	59.920 ± 1.580	74.990 ± 1.080	49.480 ± 1.200
PCDG-P (projection)	38.510 ± 0.950	14.110 ± 1.220	67.600 ± 1.860	81.540 ± 0.700	42.420 ± 1.840
PCDG-PD (+decoding)	**9.520 ± 1.040**	3.470 ± 0.640	**88.670 ± 1.740**	**93.920 ± 0.680**	**14.090 ± 2.110**
PCDG (full)	11.280 ± 1.050	4.500 ± 0.700	87.870 ± 1.590	93.460 ± 0.630	15.860 ± 1.900

Fidelity and residuals. On Istanbul, projection alone degrades Distance (0.1128 → 0.1314) and Radius (0.0226 → 0.0516), while the full stack lowers all six JSDs below JointGen. It also lowers five of six below PO-CFG; its G-RANK remains slightly higher (0.1798 vs. 0.1794). On New York, four of six JSDs improve over JointGen. CategoryTransition (0.0070 → 0.0183) and G-RANK (0.1612 → 0.1914) remain higher, and PO-CFG is best on those two and Category. The observed category-transition gap does not establish that such loss is unavoidable for partial-order control. These comparisons use three-seed means; no significance test is performed.

4.3 Ablation of the projection stage

The projection ablation removes one component at a time. Tables 3 and 4 report all six fidelity and five constraint metrics with three-seed means and standard deviations.

Removing existence lowers PairCov below the unconstrained backbone, showing that ordering alone can be satisfied by omitting constrained categories. Removing order retains coverage but raises strict OVR, separating presence from ordering. Distance KL improves the two spatial indicators in both cities, while the Gumbel and KL variants have metric-dependent effects; these are component effects rather than a claim that every term improves every fidelity statistic.

4.4 Decomposing the fidelity–constraint trade-off

Figure 3 shows the Unsat–Distance plane; the complete stage-wise means are in Table 5, Appendix C.

Projection and decoding. Decoding (b)→(c) reduces strict OVR from 32.80% to 3.72% on Istanbul and from 38.51% to 9.52% on New York. Radius increases from 0.0516 to 0.0892 and from 0.1265 to

Table 2: Ablation of the projection stage: distributional similarity (JSD ↓, evaluation v2). FULL here is PCDG-P, i.e. the projection stage alone.

Istanbul						
Variant	Distance ↓	Radius ↓	CategoryTransition ↓	DailyLoc ↓	Category ↓	G-RANK ↓
No Projection	0.1128 ± 0.0087	0.0226 ± 0.0017	0.0403 ± 0.0001	0.0414 ± 0.0009	0.0649 ± 0.0003	0.1922 ± 0.0022
No Existence	**0.1084 ± 0.0114**	**0.0215 ± 0.0027**	0.0389 ± 0.0005	0.0412 ± 0.0021	0.0661 ± 0.0009	0.1924 ± 0.0028
No Order	0.1176 ± 0.0023	0.0486 ± 0.0035	0.0227 ± 0.0008	**0.0358 ± 0.0006**	0.0439 ± 0.0018	**0.1770 ± 0.0040**
No KL	0.1406 ± 0.0117	0.0574 ± 0.0028	0.0164 ± 0.0006	0.0361 ± 0.0012	**0.0429 ± 0.0012**	0.1817 ± 0.0024
Fixed Multipliers	0.1296 ± 0.0093	0.0526 ± 0.0045	0.0185 ± 0.0005	0.0361 ± 0.0009	0.0457 ± 0.0025	0.1841 ± 0.0020
No Gumbel Noise	0.1315 ± 0.0091	0.0524 ± 0.0023	**0.0160 ± 0.0010**	0.0370 ± 0.0005	0.0439 ± 0.0023	0.1831 ± 0.0027
No Distance KL	0.1441 ± 0.0073	0.0582 ± 0.0025	0.0162 ± 0.0010	0.0366 ± 0.0008	0.0437 ± 0.0023	0.1828 ± 0.0026
FULL (=PCDG-P)	0.1314 ± 0.0035	0.0516 ± 0.0022	0.0162 ± 0.0006	0.0368 ± 0.0014	0.0446 ± 0.0020	0.1820 ± 0.0016
New York						
Variant	Distance ↓	Radius ↓	CategoryTransition ↓	DailyLoc ↓	Category ↓	G-RANK ↓
No Projection	0.2904 ± 0.0156	0.1104 ± 0.0071	**0.0070 ± 0.0008**	0.0409 ± 0.0035	0.0184 ± 0.0002	0.1612 ± 0.0013
No Existence	0.2439 ± 0.0277	**0.1004 ± 0.0053**	0.0075 ± 0.0008	0.0393 ± 0.0044	0.0189 ± 0.0004	**0.1587 ± 0.0006**
No Order	0.2175 ± 0.0284	0.1196 ± 0.0108	0.0138 ± 0.0003	0.0352 ± 0.0029	0.0177 ± 0.0003	0.1860 ± 0.0048
No KL	0.2415 ± 0.0239	0.1421 ± 0.0046	0.0125 ± 0.0006	0.0352 ± 0.0041	0.0173 ± 0.0013	0.1939 ± 0.0064
Fixed Multipliers	0.2402 ± 0.0487	0.1218 ± 0.0018	0.0127 ± 0.0005	0.0353 ± 0.0041	0.0175 ± 0.0006	0.1891 ± 0.0023
No Gumbel Noise	**0.2055 ± 0.0205**	0.1248 ± 0.0056	0.0130 ± 0.0010	0.0360 ± 0.0036	0.0173 ± 0.0003	0.1918 ± 0.0041
No Distance KL	0.2484 ± 0.0267	0.1430 ± 0.0055	0.0134 ± 0.0011	0.0355 ± 0.0031	**0.0172 ± 0.0013**	0.1937 ± 0.0082
FULL (=PCDG-P)	0.2210 ± 0.0224	0.1265 ± 0.0051	0.0132 ± 0.0004	**0.0351 ± 0.0035**	0.0173 ± 0.0013	0.1942 ± 0.0041

Mean ± sample standard deviation over three sampling seeds. Bold denotes the best unrounded mean within each city, not statistical significance. No KL disables both the token-KL anchor and the distance-KL term; NO DISTANCE KL disables only the latter. Removing the KL or Gumbel terms does not uniformly degrade fidelity, which we report rather than claiming a benefit for every component.

Table 3: Ablation of the projection stage: constraint satisfaction (%, evaluation v2). The existence energy drives coverage, the order energy drives strict ordering, and the adaptive multipliers beat a fixed penalty.

Istanbul					
Variant	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
No Projection	92.89 ± 0.08	48.05 ± 0.74	14.63 ± 0.19	43.19 ± 0.20	95.57 ± 0.15
No Existence	93.67 ± 0.15	**7.67 ± 0.56**	7.34 ± 0.12	36.92 ± 0.19	93.58 ± 0.17
No Order	63.50 ± 0.76	50.34 ± 0.73	**72.53 ± 0.70**	**86.08 ± 0.17**	70.70 ± 0.73
No KL	32.85 ± 0.52	9.20 ± 0.29	71.11 ± 0.08	85.40 ± 0.16	36.75 ± 0.57
Fixed Multipliers	42.70 ± 0.51	20.97 ± 0.90	71.25 ± 0.89	85.00 ± 0.42	45.64 ± 0.58
No Gumbel Noise	**32.75 ± 0.39**	9.13 ± 0.22	70.97 ± 0.34	85.38 ± 0.19	**36.75 ± 0.25**
No Distance KL	32.96 ± 0.46	9.10 ± 0.16	70.92 ± 0.23	85.29 ± 0.14	36.83 ± 0.24
FULL (=PCDG-P)	32.80 ± 0.50	9.12 ± 0.10	70.99 ± 0.08	85.36 ± 0.15	36.86 ± 0.37
New York					
Variant	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
No Projection	89.45 ± 0.20	52.92 ± 0.86	21.13 ± 0.15	46.75 ± 0.12	93.64 ± 0.26
No Existence	90.38 ± 0.47	18.04 ± 1.26	10.95 ± 0.45	39.47 ± 0.12	91.80 ± 0.40
No Order	65.43 ± 0.12	51.24 ± 0.30	**69.92 ± 0.74**	**82.84 ± 0.40**	72.86 ± 0.94
No KL	**38.24 ± 0.96**	13.98 ± 0.82	67.67 ± 1.56	81.57 ± 0.72	**42.40 ± 1.48**
Fixed Multipliers	58.45 ± 0.66	35.14 ± 0.60	63.79 ± 0.76	78.37 ± 0.35	60.67 ± 0.49
No Gumbel Noise	39.11 ± 1.53	13.93 ± 1.34	66.41 ± 1.32	80.95 ± 0.65	43.91 ± 1.59
No Distance KL	38.48 ± 1.04	**13.76 ± 0.72**	67.33 ± 1.86	81.38 ± 0.81	42.77 ± 1.79
FULL (=PCDG-P)	38.51 ± 0.95	14.11 ± 1.22	67.60 ± 1.86	81.54 ± 0.70	42.42 ± 1.84

Mean ± sample standard deviation over three sampling seeds. Bold denotes the best unrounded mean within each city. Removing the existence energy collapses coverage *below* the unconstrained backbone, confirming that ordering can otherwise be satisfied by omitting the constrained categories.

0.1681, respectively. The fidelity changes are metric-dependent: category-transition JSD also increases in both cities. Thus coupling the token and catalog representations strengthens realized activity-order control, but also changes the distribution of available POIs.

Geometry and distribution correction. At (c)→(d), geometry preserves constraint metrics within 10^{-12}. Distance/Radius fall from 0.1539/0.0892 to 0.0372/0.0159 on Istanbul and from 0.1989/0.1681 to 0.0610/0.0382 on New York. Because actual categories are unchanged, Distance's endpoint filter is also unchanged in this comparison. This isolates the spatial refinement effect under fixed realized activities and a fixed Distance evaluation subset. New York G-RANK nevertheless rises from 0.1914 to 0.2069. Stage (e) lowers it to 0.1914 and lowers Istanbul G-RANK from 0.1835 to 0.1798, with a modest constraint regression. This transition jointly changes ρ, β, and the candidate pool, so it does not isolate their individual effects. Selection of $\lambda = 0$ identifies a preferred configuration within

the evaluated grid, not evidence that category-side fidelity loss is absent.

Selection protocol. The 18 configurations were screened on training-side calibration records already seen by the backbone, not an independent validation set. Ranking minimizes the maximum, then mean, of twelve city–metric JSD/JointGen ratios. The two shortlisted configurations underwent three-seed confirmation; neither passed all distribution gates. The final candidate was chosen from them while retaining the constraint gates, with an explicit override of the failed distribution gates, and evaluated on the test set without reopening selection. Final ranking used v3 re-evaluation; both screening order and final candidate order agree with v2 (Appendix B). No test metric entered hyperparameter selection, candidate construction, or the refinement objective.

4.5 Cost and reproducibility

PCDG requires no retraining but adds projection, decoding, and refinement work. In the earlier geometry qualification, the selected 200-step refinement increased median spatial-sampling time by 3.35–3.98% relative to projection Full, not JointGen. These measurements do not establish overhead for the later full stack. Comparable full-stack/JointGen timing and PO-CFG training GPU-hours are not measured in the archived cost evidence. Appendix E gives measured batch times and their scope.

5 Related Work

Human mobility and POI modeling. Classical studies established that human movement is highly regular yet only boundedly predictable [13, 33]. A large body of work predicts the *next* location from history using attentional, spatio-temporal, and graph-enhanced models [9, 25, 40], and diffusion models have recently been applied to next-POI and sequential recommendation [5, 21, 28, 35, 37]. These prediction and ranking objectives differ from generating complete time-stamped visits under sample-level activity-order requirements.

Synthetic trajectory and mobility generation. Data-driven generation spans GAN/Seq2Seq models [2, 18, 20, 39], point-process and neural-ODE generators [23, 42], imitation- and privacy-oriented approaches [10, 17, 27, 29, 34, 36], and, increasingly, diffusion models [6, 12, 22, 30, 38, 41, 44] and Transformer/LLM-based generators [14, 16, 19, 43]. Marionette supports fine-grained generation using global and partial mobility contexts [7]; we use its generator as a frozen backbone. Related benchmarks examine imitative generation quality [8]. Our target is an explicit relation between activity categories, together with the distributional effects of enforcing it on generated POI identities.

Discrete diffusion. Discrete diffusion extends denoising diffusion [15] to categorical data via structured transition matrices [1], concrete score matching [26], score-entropy [24], and masked formulations [31]. Discrete classifier-free and classifier-based guidance steer generation toward conditions [32]. Our EnergyGuide baseline instead takes a gradient of the local order/existence energy; it does not reproduce that work's classifier-likelihood construction. PCDG intervenes in the frozen backbone at sampling time.

Sampling-time constraint projection. As prior background, CDD integrates differentiable constraints into discrete-diffusion sampling through KL anchoring and augmented-Lagrangian optimization without retraining [3]. PCDG builds on this general sampling-time projection idea but derives its own cross-position order energy and existence energy under a KL-anchored augmented-Lagrangian projection over category positions, rather than reusing that prior system as a dependency. The KL anchor penalizes token-level drift from the denoiser; matching aggregate travel-distance and visitation statistics to training references is a separate objective in our stack. We evaluate those terms through projection and stage-wise comparisons rather than treating the KL anchor as a distributional-fidelity guarantee.

Decoding support and spatial refinement. Grammar-constrained decoding restricts language-model outputs to permitted formal structures [11]; Control-DAG uses weighted finite-state automata for lexical, vocabulary, and length control in non-autoregressive generation [4]. Our decoder instead couples category tokens to admissible catalog POIs. Its final-category fallback prevents a universal consistency guarantee. Geographic control is complementary: ControlTraj uses road topology and trip attributes to condition trajectory generation [45]. This differs from preserving an activity sequence while changing its locations. Our terminal refinement reselects POIs within their realized categories; the stage-wise evaluation measures how much spatial fidelity this restriction permits us to recover, and which errors remain.

6 Limitations

The two-city, three-seed study does not measure training variability or statistical significance. The cities use different backbone batch sizes, and full training provenance is unavailable. PO-CFG uses scale 1 without extrapolation tuning; batch-level distance KL gives no per-sequence guarantee. Finite-budget projection and decoder fallback leave residual violations, and same-category refinement does not fix New York's category-transition gap. Training-side calibration is not independent validation; its failed distribution gates and override are detailed in §5.4 and Appendix B. Stage (d) fixes geometry settings, while (d)→(e) jointly changes three factors. Complex order DAGs, conflict detection, adaptive scheduling, training-time order losses, and recommendation utility remain open. Synthetic output has no formal privacy guarantee; distributional similarity does not establish safe release.

When the backbone assigns little mass to required categories, finite-budget projection can leave relations unsatisfied. Decoding addresses category–POI mismatch; category-preserving geometry cannot correct errors in the generated category sequence.

7 Conclusion

On a frozen generator, PCDG combines order/existence projection, category-consistent decoding, and same-category refinement to support new category-order requests without retraining. It outperforms retrained PO-CFG on all five constraint metrics in both cities and JointGen on six fidelity JSDs in Istanbul and four in New York. Refinement restores spatial fidelity while preserving constraints, but leaves New York's category-transition and popularity errors unresolved and is restricted to realized categories.

References

[1] Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. 2021. Structured Denoising Diffusion Models in Discrete State-Spaces. In *Advances in Neural Information Processing Systems*, Vol. 34. Curran Associates, Inc., 17981–17993. https://proceedings.neurips.cc/paper_files/paper/2021/hash/958c530554f78bcd8e97125b70e6973d-Abstract.html

[2] Chu Cao and Mo Li. 2021. Generating Mobility Trajectories with Retained Data Utility. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM, 2610–2620. doi:10.1145/3447548.3467158

[3] Michael Cardei, Jacob K. Christopher, Bhavya Kailkhura, Tom Hartvigsen, and Ferdinando Fioretto. 2025. Constrained Discrete Diffusion. In *Advances in Neural Information Processing Systems*, Vol. 38. Curran Associates, Inc., 12383–12416. doi:10.52202/085713-0415

[4] Jinghong Chen, Weizhe Lin, Jingbiao Mei, and Bill Byrne. 2024. ControlDAG: Constrained Decoding for Non-Autoregressive Directed Acyclic T5 using Weighted Finite State Automata. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers)*. Association for Computational Linguistics, Mexico City, Mexico, 508–518. doi:10.18653/v1/2024.naacl-short.42

[5] Jialei Chen, Yuanbo Xu, and Yiheng Jiang. 2025. Unlocking the Power of Diffusion Models in Sequential Recommendation: A Simple and Effective Approach. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*. ACM, 155–166. doi:10.1145/3711896.3737172

[6] Chen Chu, Hengcai Zhang, Peixiao Wang, and Feng Lu. 2024. Simulating Human Mobility with a Trajectory Generation Framework Based on Diffusion Model. *International Journal of Geographical Information Science* 38, 5 (2024), 847–878. doi:10.1080/13658816.2024.2312199

[7] Bangchao Deng, Ling Ding, Lianhua Ji, Chunhua Chen, Xin Jing, Bingqing Qu, and Dingqi Yang. 2025. Marionette: Fine-Grained Conditional Generative Modeling of Spatiotemporal Human Trajectory Data Beyond Imitation. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*. ACM, 463–473. doi:10.1145/3711896.3737039

[8] Bangchao Deng, Xin Jing, Tianyue Yang, Bingqing Qu, Dingqi Yang, and Philippe Cudre-Mauroux. 2025. Revisiting Synthetic Human Trajectories: Imitative Generation and Benchmarks Beyond Datasaurus. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.1*. ACM, 201–212. doi:10.1145/3690624.3709180

[9] Jie Feng, Yong Li, Chao Zhang, Funing Sun, Fanchao Meng, Ang Guo, and Depeng Jin. 2018. DeepMove: Predicting Human Mobility with Attentional Recurrent Networks. In *Proceedings of the 2018 World Wide Web Conference*. ACM, 1459–1468. doi:10.1145/3178876.3186058

[10] Jie Feng, Zeyu Yang, Fengli Xu, Haisu Yu, Mudan Wang, and Yong Li. 2020. Learning to Simulate Human Mobility. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, 3426–3433. doi:10.1145/3394486.3412862

[11] Saibo Geng, Martin Josifoski, Maxime Peyrard, and Robert West. 2023. Grammar-Constrained Decoding for Structured NLP Tasks without Finetuning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Singapore, 10932–10952. doi:10.18653/v1/2023.emnlp-main.674

[12] Letian Gong, Shengnan Guo, Yan Lin, Yichen Liu, Erwen Zheng, Yiwei Shuang, Youfang Lin, Jilin Hu, and Huaiyu Wan. 2025. STCDM: Spatio-Temporal Contrastive Diffusion Model for Check-In Sequence Generation. *IEEE Transactions on Knowledge and Data Engineering* 37, 4 (2025), 2141–2154. doi:10.1109/TKDE.2025.3525718

[13] Marta C. González, César A. Hidalgo, and Albert-László Barabási. 2008. Understanding Individual Human Mobility Patterns. *Nature* 453, 7196 (2008), 779–782. doi:10.1038/nature06958

[14] Ammar Haydari, Dongjie Chen, Zhengfeng Lai, Michael Zhang, and Chen-Nee Chuah. 2026. MobilityGPT: Enhanced Human Mobility Modeling With a GPT Model. *IEEE Transactions on Intelligent Transportation Systems* 27, 1 (2026), 1681–1694. doi:10.1109/TITS.2025.3626357

[15] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising Diffusion Probabilistic Models. In *Advances in Neural Information Processing Systems*, Vol. 33. Curran Associates, Inc., 6840–6851. https://proceedings.neurips.cc/paper_files/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html

[16] Shang-Ling Hsu, Emmanuel Tung, John Krumm, Cyrus Shahabi, and Khurram Shafique. 2024. TrajGPT: Controlled Synthetic Trajectory Generation Using a Multitask Transformer-Based Spatiotemporal Model. In *Proceedings of the 32nd ACM International Conference on Advances in Geographic Information Systems*. ACM, 362–371. doi:10.1145/3678717.3691303

[17] Yujia Hu, Yuntao Du, Zhikun Zhang, Ziquan Fang, Lu Chen, Kai Zheng, and Yunjun Gao. 2024. Real-Time Trajectory Synthesis with Local Differential Privacy. In *Proceedings of the IEEE International Conference on Data Engineering*. IEEE, 1685–1698. doi:10.1109/ICDE60146.2024.00137

[18] Wenjun Jiang, Wayne Xin Zhao, Jingyuan Wang, and Jiawei Jiang. 2023. Continuous Trajectory Generation Based on Two-Stage GAN. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. Association for the Advancement of Artificial Intelligence (AAAI), 4374–4382. doi:10.1609/aaai.v37i4.25557

[19] Siyu Li, Toan Tran, Haowen Lin, John Krumm, Cyrus Shahabi, Lingyi Zhao, Khurram Shafique, and Li Xiong. 2025. Geo-Llama: Leveraging LLMs for Human Mobility Trajectory Generation with Constraints. In *Proceedings of the 2025 IEEE International Conference on Mobile Data Management*. IEEE, 20–31. doi:10.1109/MDM65600.2025.00023

[20] Haowen Lin, Sina Shaham, Yao-Yi Chiang, and Cyrus Shahabi. 2023. Generating Realistic and Representative Trajectories with Mobility Behavior Clustering. In *Proceedings of the 31st ACM International Conference on Advances in Geographic Information Systems*. ACM, 1–4. doi:10.1145/3589132.3625657

[21] Jing Long, Guanhua Ye, Tong Chen, Yang Wang, Meng Wang, and Hongzhi Yin. 2024. Diffusion-Based Cloud-Edge-Device Collaborative Learning for Next POI Recommendations. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM, 2026–2036. doi:10.1145/3637528.3671743

[22] Qingyue Long, Can Rong, Tong Li, and Yong Li. 2026. Dynamic Population Distribution Aware Human Trajectory Generation with Diffusion Model. *ACM Transactions on Intelligent Systems and Technology* 17, 3 (2026), 1–23. doi:10.1145/3787459

[23] Qingyue Long, Huandong Wang, Tong Li, Lisi Huang, Kun Wang, Qiong Wu, Guangyu Li, Yanping Liang, Li Yu, and Yong Li. 2023. Practical Synthetic Human Trajectories Generation Based on Variational Point Processes. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM, 4561–4571. doi:10.1145/3580305.3599888

[24] Aaron Lou, Chenlin Meng, and Stefano Ermon. 2024. Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution. In *Proceedings of the 41st International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 235)*. PMLR, 32819–32848. https://proceedings.mlr.press/v235/lou24a.html

[25] Yingtao Luo, Qiang Liu, and Zhaocheng Liu. 2021. STAN: Spatio-Temporal Attention Network for Next Location Recommendation. In *Proceedings of the Web Conference 2021*. ACM, 2177–2185. doi:10.1145/3442381.3449998

[26] Chenlin Meng, Kristy Choi, Jiaming Song, and Stefano Ermon. 2022. Concrete Score Matching: Generalized Score Matching for Discrete Data. In *Advances in Neural Information Processing Systems*, Vol. 35. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 34532–34545. doi:10.52202/068431-2502

[27] Fabian Netzler and Markus Lienkamp. 2024. Privacy Preserving Human Mobility Generation Using Grid-Based Data and Graph Autoencoders. *ISPRS International Journal of Geo-Information* 13, 7 (2024), 245. doi:10.3390/ijgi13070245

[28] Yifang Qin, Hongjun Wu, Wei Ju, Xiao Luo, and Ming Zhang. 2024. A Diffusion Model for POI Recommendation. *ACM Transactions on Information Systems* 42, 2 (2024), 1–27. doi:10.1145/3624475

[29] Jinmeng Rao, Song Gao, and Sijia Zhu. 2023. CATS: Conditional Adversarial Trajectory Synthesis for Privacy-Preserving Trajectory Data Publication Using Deep Learning Approaches. *International Journal of Geographical Information Science* 37, 12 (2023), 2538–2574. doi:10.1080/13658816.2023.2262550

[30] Xuan Rao, Shuo Shang, Renhe Jiang, Peng Han, and Lisi Chen. 2025. Seed: Bridging Sequence and Diffusion Models for Road Trajectory Generation. In *Proceedings of the ACM Web Conference 2025*. ACM, 2007–2017. doi:10.1145/3696410.3714951

[31] Subham Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin Chiu, Alexander Rush, and Volodymyr Kuleshov. 2024. Simple and Effective Masked Diffusion Language Models. In *Advances in Neural Information Processing Systems*, Vol. 37. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 130136–130184. doi:10.52202/079017-4135

[32] Yair Schiff, Subham Sekhar Sahoo, Hao Phung, Guanghan Wang, Sam Boshar, Hugo Dalla-torre, Bernardo P. de Almeida, Alexander M. Rush, Thomas Pierrot, and Volodymyr Kuleshov. 2025. Simple Guidance Mechanisms for Discrete Diffusion Models. In *The Thirteenth International Conference on Learning Representations*. https://openreview.net/forum?id=i5MrJ6g5G1

[33] Chaoming Song, Zehui Qu, Nicholas Blumm, and Albert-László Barabási. 2010. Limits of Predictability in Human Mobility. *Science* 327, 5968 (2010), 1018–1021. doi:10.1126/science.1177170

[34] Xinyue Sun, Qingqing Ye, Haibo Hu, Yuandong Wang, Kai Huang, Tianyu Wo, and Jie Xu. 2023. Synthesizing Realistic Trajectory Data With Differential Privacy. *IEEE Transactions on Intelligent Transportation Systems* 24, 5 (2023), 5502–5515. doi:10.1109/TITS.2023.3241290

[35] Dongsheng Wang, Yuxi Huang, Shen Gao, Yifan Wang, Chengrui Huang, and Shuo Shang. 2025. Generative Next POI Recommendation with Semantic ID. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*. ACM, 2904–2914. doi:10.1145/3711896.3736981

[36] Huandong Wang, Changzheng Gao, Yuchen Wu, Depeng Jin, Lina Yao, and Yong Li. 2023. PateGail: A Privacy-Preserving Mobility Trajectory Generator with Imitation Learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. Association for the Advancement of Artificial Intelligence (AAAI), 14539–14547. doi:10.1609/aaai.v37i12.26700

[37] Ziwei Wang, Jun Zeng, Lin Zhong, Ling Liu, Min Gao, and Junhao Wen. 2024. DS-DRec: Next POI Recommendation Using Deep Semantic Extraction and Diffusion Model. *Information Sciences* 678 (2024), 121004. doi:10.1016/j.ins.2024.121004

[38] Tonglong Wei, Youfang Lin, Shengnan Guo, Yan Lin, Yiheng Huang, Chenyang Xiang, Yuqing Bai, and Huaiyu Wan. 2024. Diff-RNTraj: A Structure-Aware Diffusion Model for Road Network-Constrained Trajectory Generation. *IEEE Transactions on Knowledge and Data Engineering* 36, 12 (2024), 7940–7953. doi:10.1109/TKDE.2024.3460051

[39] Gang Xiong, Zhishuai Li, Meihua Zhao, Yu Zhang, Qinghai Miao, Yisheng Lv, and Fei-Yue Wang. 2024. TrajSGAN: A Semantic-Guiding Adversarial Network for Urban Trajectory Generation. *IEEE Transactions on Computational Social Systems* 11, 2 (2024), 1733–1743. doi:10.1109/TCSS.2023.3235923

[40] Song Yang, Jiamou Liu, and Kaiqi Zhao. 2022. GETNext: Trajectory Flow Map Enhanced Transformer for Next POI Recommendation. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. ACM, 1144–1153. doi:10.1145/3477495.3531983

[41] Yuan Yuan, Jingtao Ding, Chenyang Shao, Depeng Jin, and Yong Li. 2023. Spatio-temporal Diffusion Point Processes. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM, 3173–3184. doi:10.1145/3580305.3599511

[42] Yuan Yuan, Jingtao Ding, Huandong Wang, Depeng Jin, and Yong Li. 2022. Activity Trajectory Generation via Modeling Spatiotemporal Dynamics. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM, 4752–4762. doi:10.1145/3534678.3542671

[43] Kunyi Zhang, Yanbo Pang, Yurong Zhang, and Yoshihide Sekimoto. 2024. MobGLM: A Large Language Model for Synthetic Human Mobility Generation. In *Proceedings of the 32nd ACM International Conference on Advances in Geographic Information Systems*. ACM, 629–632. doi:10.1145/3678717.3691311

[44] Yuanshao Zhu, Yongchao Ye, Shiyao Zhang, Xiangyu Zhao, and James J. Q. Yu. 2023. DiffTraj: Generating GPS Trajectory with Diffusion Probabilistic Model. In *Advances in Neural Information Processing Systems*, Vol. 36. Curran Associates, Inc., 65168–65188. doi:10.52202/075280-2843

[45] Yuanshao Zhu, James Jianqiao Yu, Xiangyu Zhao, Qidong Liu, Yongchao Ye, Wei Chen, Zijian Zhang, Xuetao Wei, and Yuxuan Liang. 2024. ControlTraj: Controllable Trajectory Generation with Topology-Constrained Diffusion Model. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM, 4676–4687. doi:10.1145/3637528.3671866

A Additional Implementation Settings

Algorithm 1 and §5.1 give the controlled step and main sampling settings. Dual variables initialize at 1, penalties grow by 2 up to 10^3, and the violation threshold is $\tau = 0$ with multiplier-update tolerance 10^{-6}. Early stopping is disabled. In the batch implementation, token KL sums over vocabulary and positions for active sequences and divides by the full batch size; order/existence penalties sum over active relations and sequences without that division. Geometry uses distance/radius/fidelity weights $(1, 2, 0.1)$, 8 paths, up to 32 candidates, temperature 1, Adam rate 0.05, 200 steps, and noise refresh every 20 steps. The full-stack weights are $(\rho, \lambda, \beta) = (0.5, 0, 1)$. Geometry's timing qualification is in Appendix E.

B Distribution-Correction Grid and Selection

We screened $\rho \in \{0.25, 0.5, 1\}$, $\lambda \in \{0, 1\}$, and $\beta \in \{0, 0.1, 1\}$ on 512 training-side conditions per city with one sampling seed. Table 2 retains the archived *v2* screening values. Screening requires all five constraint metrics to be no worse than PCDG-P in each city: strict OVR, skip OVR and Unsat must not increase, and pair/category coverage must not decrease. Confirmation additionally requires each of the six JSDs to be no greater than JointGen's. Comparisons use unrounded city-wise sampling-seed means, an absolute tolerance of 10^{-12}, and reject undefined metrics. Confirmation uses a separate 512-condition training-side pool per city; the remaining training records form the distribution-correction reference pool. The shared-city ranking minimizes the maximum, then mean, of twelve JSD/JointGen ratios, followed by the parameter tuple to break ties. Upstream samples are cached by city, calibration pool, seed, ρ, and λ, so the three β refinements share upstream randomness and the grid uses a fixed candidate-pool construction.

Table 4: Training-side screening (v2): maximum and mean of twelve city–metric JSD/JointGen ratios. * denotes the two candidates sent to three-seed confirmation, not passage of its distribution gates.

Rank	ρ	λ	β	Max	Mean
1*	0.5	0	1	1.7121	0.9148
2*	0.5	0	0.1	1.7121	0.9540
3	0.5	0	0	1.7121	0.9725
4	0.25	0	0	1.7250	0.9477
5	0.25	0	0.1	1.7250	0.9534
6	0.25	0	1	1.7250	0.9604
7	0.5	1	1	1.7515	0.9599
8	0.5	1	0.1	1.7515	0.9739
9	0.5	1	0	1.7515	0.9817
10	0.25	1	0	1.7619	0.9594
11	0.25	1	0.1	1.7619	0.9695
12	0.25	1	1	1.7619	0.9782
13	1	1	1	1.7805	0.9578
14	1	1	0.1	1.7805	0.9951
15	1	1	0	1.7805	1.0046
16	1	0	1	1.7841	0.9501
17	1	0	0.1	1.7841	0.9962
18	1	0	0	1.7841	0.9989

The top two configurations, $(0.5, 0, 1)$ and $(0.5, 0, 0.1)$, underwent three-seed confirmation. Both retained the constraint gates and met the Distance, Radius, and DailyLoc ceilings, but failed CategoryTransition, Category, and G-RANK ceilings on calibration data. Thus the original all-gates confirmation selected no candidate. Formal sampling subsequently used an explicit override: rank the two confirmed candidates while retaining constraint gates but allowing failed distribution gates.

This final ranking used the archived *v3* deterministic-G-RANK re-evaluation. The 18 screening ranks, two-candidate shortlist, failed confirmation gates, and final candidate order are unchanged under v2. For the selected $(0.5, 0, 1)$, the maximum/mean confirmation ratios are 1.4628/0.8884 in v3 and 1.4628/0.8909 in v2; the runner-up means are 0.9103 and 0.9135, respectively. The selected configuration was frozen before formal test sampling. Test metrics were used only for reporting: eleven of eleven gates hold on Istanbul and nine on New York. The calibration pool was seen during backbone training and is not an independent validation set.

C Stage-wise Tables

Table 5 reports the full stage-wise comparison. All values follow evaluation v2.

D Fidelity Metrics and Audit Details

Section **??** defines the five catalog-category constraint metrics. The separate diagnostic `token_strict_ovr` uses category tokens. Mismatch rate pools the event-wise disagreements between tokens and catalog categories.

JSD and filtering. For count vectors P, Q, the implementation forms $p = P/\sum P + \epsilon$, $q = Q/\sum Q + \epsilon$, $\epsilon = 10^{-14}$, without a second normalization, and evaluates $\frac{1}{2}\sum_j p_j \log(p_j/m_j) + \frac{1}{2}\sum_j q_j \log(q_j/m_j)$, $m = (p + q)/2$, using natural logarithms. A zero-total vector is undefined. Distance, Radius, DailyLoc, and G-RANK retain sequences

Table 5: Stage-wise comparison, evaluation-v2 three-seed means. SDs for (a), (b), (c), and (e) are in Table 1. Geometry (d) preserves (c)'s constraints within 10^{-12} and uses a previously fixed configuration. Transition (d)→(e) jointly changes ρ $(1 \to 0.5)$, β $(0 \to 1)$, and the same-category candidate pool; it does not isolate a single factor. The calibration grid (Appendix B) holds the pool construction fixed.

	Distributional similarity (JSD ↓)						Constraints (%)		
Stage	Distance	Radius	CatTrans	DailyLoc	Category	G-RANK	OVR_str. ↓	Pair cov. ↑	Unsat ↓
Istanbul									
(a) JointGen (no control)	0.1128	0.0226	0.0403	0.0414	0.0649	0.1922	92.89	14.63	95.57
(b) + order/existence projection	0.1314	0.0516	0.0162	0.0368	0.0446	0.1820	32.80	70.99	36.86
(c) + category-consistent decoding	0.1539	0.0892	0.0238	0.0297	0.0423	0.1813	3.72	94.74	6.90
(d) + geometry refinement	0.0372	0.0159	0.0238	0.0284	0.0423	0.1835	3.72	94.74	6.90
(e) + distribution correction (full)	0.0395	0.0166	0.0235	0.0249	0.0423	0.1798	4.04	94.68	7.29
New York									
(a) JointGen (no control)	0.2904	0.1104	0.0070	0.0409	0.0184	0.1612	89.45	21.13	93.64
(b) + order/existence projection	0.2210	0.1265	0.0132	0.0351	0.0173	0.1942	38.51	67.60	42.42
(c) + category-consistent decoding	0.1989	0.1681	0.0182	0.0335	0.0178	0.1914	9.52	88.67	14.09
(d) + geometry refinement	0.0610	0.0382	0.0182	0.0335	0.0178	0.2069	9.52	88.67	14.09
(e) + distribution correction (full)	0.0698	0.0373	0.0183	0.0301	0.0175	0.1914	11.28	87.87	15.86

with at least two GPS points, independently on each side; the two category metrics include shorter sequences. Distance additionally requires the aligned generated/reference first and last categories to match. This subset can vary by method, so the reported Distance is not an unconditional whole-population divergence.

Six fidelity statistics. Distance is cumulative consecutive-point Haversine distance (Earth radius 6371 km). With arithmetic latitude/longitude centroid $\bar{x}$, Radius is

$$\text{Radius} = \sqrt{n^{-1} \sum_j d_{\text{Hav}}(x_j, \bar{x})}.$$

Its unit is $\sqrt{\text{km}}$ and it is *not* root-mean-square radius of gyration. DailyLoc counts distinct POI identities per sequence. For these three histograms, edges are exactly $\text{arange}(0, M, M/100)$, where M is the larger observed maximum. This usually produces 99 bins, omits M, drops values beyond the last edge, and renormalizes retained mass. If $M = 0$, the divergence is 0; a zero histogram otherwise yields an undefined value. These historical conventions are preserved for comparability, not proposed as new estimators.

Category averages category-frequency JSDs equally over the 24 hours. CategoryTransition is the equal-weight mean of conditional next-category JSDs over source categories with any outgoing transition on either side; self-transitions are included and sequence boundaries excluded. A row/hour empty on both sides contributes 0; one empty on only one side contributes $\ln 2$. No events or no transitions, respectively, gives an undefined metric. G-RANK retains each side's top-1000 visited POIs (all if fewer), aligns their union by POI identity with zeros for missing retained IDs, and computes JSD after normalizing retained mass. It compares identities, not rank positions.

Deterministic audit. Evaluation v2 uses an unstable frequency sort at the top-1000 cutoff. The v3 audit breaks ties by POI id; identity alignment is present in both versions. Across the 18 formal audit metric sets (six full-stack outputs and twelve reused baseline sets), the largest absolute G-RANK change is 0.0032; other statistics agree within 10^{-12}. The full-stack/backbone G-RANK comparison remains $0.1766 < 0.1892$ on Istanbul and $0.1912 > 0.1628$ on New York under v3. This audit does not replace any v2 table cell.

Table 6: Selected 200-step geometry qualification. Median seconds per 64-sequence spatial batch; ratio = Full+Geo / projection Full.

City	Batch	Full	Full+Geo	Ratio
NY	ordinary	27.028	28.063	1.0383
NY	long	33.102	34.211	1.0335
IST	ordinary	24.865	25.856	1.0398
IST	long	27.399	28.371	1.0355

E Cost

Table 6 reports the earlier selected-geometry qualification, not later full-stack timing. For each city, ordinary and long cases are eligible 64-sequence batches selected by lower-median and maximum batchwise generated length. Each uses two warmups, five measured repetitions, seed 135398, FP32 with TF32 disabled, and the same CUDA device and input. CUDA synchronization brackets the timer; ratios divide medians rather than averaging per-trial ratios.

The timer covers spatial sampling and decoding, including fresh scoring, candidate construction, optimization, and discrete selection with geometry; it excludes temporal generation, model loading, and transfer. The denominator is projection Full (PCDG-P), not JointGen. The selected 200-step configuration uses radius/fidelity weights 2/0.1; an earlier step-budget probe used 1/0.01. All four ratios are below the 1.10 ceiling, with overhead 3.35–3.98%. These measurements do not transfer to PCDG-PD plus geometry or the final $(0.5, 0, 1)$ stack. Comparable full-stack/JointGen sampling overhead and PO-CFG training GPU-hours remain unmeasured in the archived cost evidence.

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.